Mehdi Eskandari-Ghadi

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EDUCATION

Civil Engineering Ph.D. (GPA 4.0) — **Boulder, CO**
Engineering Science Focus — 2016 – 2021
University of Colorado Boulder

Thesis: "Effect of surface forces on the mechanics of adsorption-deformation in microporous media and in subcritical crack growth in reactive environment"
Advisor: Yida Zhang, Ph.D.
Area of study: Surface poromechanics, adsorption-deformation, fracture mechanics

Bachelor of Science — Civil Engineering (GPA 18.41/20) — **Tehran, Iran**
University of Tehran — 2012 – 2016
Ranked 5th among 113 undergraduate civil engineering students of the same entrance
Average GPA of all students at the Civil Engineering Department: 15.02/20

High School — Tehran, Iran
Allameh Helli 3 (NODET) High School — 2009 – 2012
Total GPA: 19.98/20

SELECTED COURSEWORK

Continuum Mechanics	A
Computational Fluid Dynamics (CFD)	A
Finite Element Method (FEM)	A
Partial Differential & Integral Equations	A
Wave-based Methods and Applications	A
Fourier Series & Boundary Value Problems	A
Complex Variables	A
Soil Dynamics	A
Soil Mechanics	A
Structural Analysis II	20/20
Structural Analysis I	19.14/20
Mechanics of Materials II	18/20
Mechanics of Materials I	18.5/20
Statics	19.9/20
Ordinary Differential Equations	20/20
Calculus II	20/20
Calculus I	20/20

HONORS AND CERTIFICATES

- Engineering Mechanics Institute (EMI) Conference award: Runner up in the 2021 Poromechanics Committee Student Poster Competition
- The Doctoral Assistantship for Excellence from University of Colorado, Boulder
- Honorary Admission for Graduate Study from the University of Tehran
- E.I.T. — Civil FE passed
- Ranked 183rd among approximately 250,000 participants in Iran's 2011 university entrance exam (Konkour)
- Ranked among the top 10 percent in every semester during Bachelor's study
- Top Researcher award in Allameh Helli 3 Seminar of Science and Technology
- Member of the Secretariat Committee of the Allameh Helli 3 Seminar of Science and Technology
- Certificate of Participation in RoboCup IRAN OPEN 2010 Competitions and Symposium
- Certificate of Education in elementary MATLAB programming
- TOEFL: 115/120 (Reading: 29, Listening: 30, Speaking: 28, Writing: 28)
- GRE: Verbal 155/170, Quantitative 170/170, Analytical Writing 4.0/6.0

ENGINEERING EXPERIENCE

Reliability & Standards Software Engineer II **Jan 2025 – Present**
PacifiCorp

- Applied full-stack TypeScript and Python engineering to design and implement the Palantir Foundry platform as the company's production solution, migrating prototype FERC-881 line-rating and FAC-008 equipment-rating business logic into an integrated system covering audit preparation, data management UIs, interdepartmental data collection, and queue-based processing pipelines.
- Built a Foundry-based ecosystem for PacifiCorp, spanning an Ontology with 100+ Actions and Functions and 30+ Pipeline Builder pipelines, to manage transmission facility and equipment records, track and approve record modifications, and compute granular, operations-scale rating calculations, including external system interfacing and efficient data management workflows delivered through automations, map interfaces, graphs, and other data solution UI elements; owned end-to-end UX design for the system - including all interaction and workflow design via Foundry Workshop - continuously identifying friction points and iterating on layout and information architecture to keep complex, multi-source engineering workflows simple and intuitive for end users.
- Directed AI coding agents with repository and Foundry Ontology access to write and edit functions and entire features - including large formatted Excel report generation, large-scale dataframe operations logic, and React-based custom Workshop widgets - through a full commit, tag, and pull-request workflow; coordinated agent-driven changes across multiple interdependent repositories that import one another as libraries, maintaining version and compatibility consistency across them.
- Built a reusable Python report-processing library supporting multiple internal and external data exports, and used it to power an LLM-based reporting pipeline via Palantir AIP - traversing Foundry Ontology object sets into dataframes, formatting outputs into schema-versioned, typed data structures with a matching data legend, and applying system, task, and report-format prompts to generate narrative reports on facility modification changes and their rating implications for administrative review and approval prior to publishing.
- Applied git-inspired version-control concepts for system analysis lifecycle management within Foundry, supporting cloning, modification, and staged publication of facility analyses through department-specific review checkpoints, with automated resource tracking and validation.
- Leveraging Spark and Polars, designed backend Python data pipelines for large-scale ingestion, parsing, and health remediation of legacy spreadsheets, automating data cleanup and multi-source merging to bring records to quality standards.
- Utilized knowledge of physics and numerical methods to build a reusable Python rating-calculation library implementing IEEE-compliant thermal rating models, solving the nonlinear ordinary-differential heat balance (solar heat gain, convective cooling, radiative heat loss, conductor resistance) through Runge-Kutta forward integration and Newton-Raphson inverse root-finding, vectorized at scale for static, dynamic, and transient line and equipment ratings within the platform.
- Applied VBA/Excel programming skills alongside IEEE equipment-rating knowledge to design and build a streamlined, user-friendly FAC-008 substation equipment rating tool from scratch, computing normal and emergency ratings and ranking limiting elements for substation equipment - including breakers, switches, capacitors, reactors, buses, jumpers, and transformers - from company standards and nameplate specifications.
- Applied geometric and spatial analysis methods to compute mutual-impedance candidate pairs, line-distance statistics, and line-to-line distance visualizations for parallel transmission corridors, integrated into platform UI graphs.
- Deployed AutoCAD experience and design architecture skills to implement AutoLISP one-line-drawing functionality to parse shape geometries and layer data - identifying transmission facilities and equipment, deducing line segment and equipment configurations, and exporting structured data to reconcile database records with engineering drawings.

Transmission Engineer II **Feb 2024 – Jan 2025**
PacifiCorp

- Utilizing high-level knowledge of transmission line design to scope new transmission lines in collaboration with other engineering disciplines.
- Initiating process improvement pipelines for transmission design processes and documentations.
- Performing detail-oriented technical review of transmission standards and collaboration in their developments.
- Continued all duties from Transmission Engineer I position.

Transmission Engineer I **Jan 2023 – Feb 2024**
PacifiCorp

- Utilizing in-depth knowledge of PLSCADD, L-PILE, NESC, and GO-95 for design of new power transmission lines and rebuilding/maintenance of existing lines in states of UT, WY, OR, WA, and CA in collaboration with teams of engineers, managers, field crew, and legal.
- Utilizing in-depth knowledge of structural engineering and geotechnical engineering for custom-engineered solutions of

transmission structures and drilled pier foundations with reference to ASCE, ACI, and FHWA to meet special needs.
- Overhauled the company's transmission structure library from legacy templates into modular, component-based PLS-POLE libraries, and designed a new class of circuit-swap structures for repositioning circuits on a double-circuit extra-high-voltage line with minimal increase to structure size, vetted across engineering and field operations to meet electrical and structural requirements.
- Utilized VBA and Excel programming alongside transmission line rating engineering knowledge to design and build an auditable, automated FERC-881 steady-state line-rating analysis tool from scratch - integrating PLS-CADD structure data, conductor properties, and historical weather data with CIGRE-referenced seasonal rating methodology to compute normal and emergency ratings and critical-span summaries - streamlining grid line data extraction, reducing engineering hours threefold and costs by approx. $1.5M, and initializing the Foundry datasets that the system later grew from.
- Employed excel programming and data analysis skills to develop automated tools for deep foundation design, construction package generation, and transmission pole design aids.
- Utilized transmission engineering knowledge and programming experience to generate company sag-tension standards for tree-wire.
- Improved company standards and company procedures with detail-oriented reviews/QAQC.
- Designing and reviewing drawings for tubular steel poles and drilled piers to withstand vertical and lateral loads.

Self-motivated Projects July 2024 – Jan 2025

- Designed and developed a full-stack web app tool for project management, resource management, and business analysis using Flask, Django, JavaScript, HTML, SQL, and third-party API integration.
- Created software for transforming PLS-CADD transmission models into line rating analysis graph-based and tree-based models using Python, OOP, and design patterns.
- Developed a calculation tool for the bending capacity of hollow polygonal tapered poles by applying structural engineering skills and Python.

Structural Engineering/Construction Intern **May 2015 – Aug 2015**

Kayson Construction Co., Tehran, Iran

- Worked with a team of 15+ engineers, 20+ construction managers, and 50+ construction workers.
- Performed daily construction quality and safety inspections.
- Prepared 30+ quality and safety inspection reports.

AutoCADman and Structural Engineering Intern **May 2014 – Aug 2014**

Sazeh Pardazi Iran Consultant Co., Tehran, Iran

- Prepared 30+ sheets of technical drawings for concrete and steel structures, including beams, columns, foundations, retaining walls, slabs, and joint details.
- Apprenticed with a team of civil engineers and architects for design of steel and reinforced concrete beams, columns, and foundations.

Class Projects **Dec 2015 and May 2016**

- Utilized ETABS for the design of a 5-story residential reinforced concrete structure and steel structure under standard combinations of live, dead, wind, and seismic loads and verified by hand calculations.
- Utilized Plaxis and GeoStudio for the senior design project of retaining wall stability with and without nailing.

RESEARCH EXPERIENCE

Graduate Research Assistant / Ph.D. Candidate / Postdoctoral Researcher **Jun 2018 – Jan 2023**

University of Colorado Boulder, Boulder, CO

- Collaborated with a team of three senior researchers at the Lawrence Berkeley National Laboratory to investigate and develop mechanistic models for subcritical crack growth and crack healing, with reference to 400+ research and technical articles.
- Investigated subcritical crack growth, crack healing, experimental apparatus, and utilization of optical birefringence techniques in fracture analysis with collaborators at LBNL.
- Utilized knowledge of fundamentals of fracture mechanics, fluid transport, and boundary element methods to develop a software that couples fields of mechanics and physics.
- Investigated the theory and engineering application of multiphysical and multiscale solid-fluid interactions (e.g., adsorption, surface tension, solvation pressure) in micro/meso/macroporous solids.
- Developed, implemented, and validated a poromechanical model for adsorption-induced deformation of microporous material in MATLAB, based on thermodynamics, poromechanics, and surface mechanics.

- Employed technical writing and illustrative skills for preparation of 4 peer-reviewed publications in journals of JMPS, IJSS, and EFM.

Research and Implementation of granular media constitutive models **2016 – 2018**
University of Colorado Boulder, Boulder, CO

- Research and implementation of elasto-plastic hardening constitutive models for modeling of sands in geomechanics.
- Implemented granular material constitutive models in ABAQUS to investigate the response of dense and loose sands under triaxial loading and generate failure surface plots.
- Programmed a discrete element MATLAB code for modeling of grain structure stability and avalanche.

Design and Setup of Shaking Table **2009 – 2010**
Allameh Helli 3 High School, Tehran, Iran (Highschool sophomore project)

- Designed and built a shaking table for evaluation of the natural frequencies of continuous cantilever beams.

Design and Building of Line-Seeker Rescue Robot **2009 – 2010**
Allameh Helli 3 High School, Tehran, Iran (Highschool junior project)

- Designed and built a line-seeker rescue robot for participation in the RoboCup IRAN OPEN 2010 competitions.

Huffman Code Research and Implementation **2008 – 2009**
Allameh Helli 3 High School, Tehran, Iran (Highschool freshman project)

- Researched and implemented the Huffman Code Theory for loss-less data encoding and file compression.

PUBLICATIONS

Journal Articles

- Eskandari-Ghadi, Mehdi, and Yida Zhang. "Mechanics of shrinkage-swelling transition of microporous materials at the initial stage of adsorption." *International Journal of Solids and Structures* **222** (2021): 111041.
- Mehdi Eskandari-Ghadi and Yida Zhang. "Effect of pore size distribution on sorption-induced deformation of porous materials: a theoretical study." *International Journal of Solids and Structures* (2022), published online.
- Mehdi Eskandari-Ghadi, Deng Hang, Seiji Nakagawa, Steve Pride, Benjamin Gilbert, Yida Zhang. "Surface force and reactant transport as the central mechanisms for subcritical crack growth: a theoretical study." *Journal of Mechanics and Physics of Solids* (2022).
- Seiji Nakagawa, Yida Zhang, Mehdi Eskandari-Ghadi, Donald W. Vasco. "Corrections of Double-Torsion (DT) subcritical crack growth tests for crack profile geometry." *Engineering Fracture Mechanics* (2022).

CONFERENCE TALKS AND PRESENTATIONS

EMI Virtual Conference 2021 and EMI Conference 2022 **May 2021 and May 2022**

- Applied illustrative and verbal communication skills to deliver presentations titled "A multi-scale theory explaining the initial shrinkage of micro-porous solid upon gas adsorption" and "A Multiphysical Surface-Force Based Fracture Theory for Subcritical Crack Growth in Surface-Reactive Environments", followed by Q&A from engineering and academic audience.
- Received runner up award in the 2021 Student Poster Competition for poster presentation on sorption-induced deformation of porous material.

Lawrence Berkeley National Laboratory BES — Regular Presentations **Sep 2020 – Jan 2023**

- Presented regular scientific presentations on topic of crack propagation to teams of research professionals and senior scientists in the LBNL BES with diverse backgrounds in mechanics, physics, and chemistry.
- Presented a 30-minute abstract for senior scientists in the LBNL BES annual meeting on the effects of surface forces in subcritical crack propagation.

TEACHING EXPERIENCE

Advanced Mechanics of Materials (Graduate level) — Teaching Assistant **Fall 2016, Fall 2017**
University of Colorado Boulder, Boulder, CO

- Responsible for holding office hours, preparation of homework and exam questions, and grading of homework, lab reports, and exams.
- Held 9 laboratory teaching sessions for measurement of elastic properties of materials using extension, 3-point bending, and torsion experiments.

Analytical Mechanics II — Teaching Assistant **Spring 2017**

University of Colorado Boulder, Boulder, CO

- Responsible for holding office hours, preparation of homework and exam questions, and grading of homework.

Dynamics (Undergraduate level) — Teaching Assistant **2018 – 2019**
University of Colorado Boulder, Boulder, CO

- Presented 3 class lectures, held 190 hours of office hours and exam preparation sessions, prepared and graded 6 exams and 45 homework question sets.
- Tailored technical communication specific to each student's understanding and utilized AutoCAD 3D drawings and cross sections to enhance visualization and understanding.

FE Exam Tutoring

- Tutored students for FE exam with 100% success rate.

Structural Analysis II — Teaching Assistant **Fall 2014**
University of Tehran, Tehran, Iran

- Responsible for grading homework and class projects.

Hydraulics of Open Channels — Teaching Assistant **Fall 2015**
University of Tehran, Tehran, Iran

- Responsible for holding office hours, preparation of homework and quizzes, and grading of homework and quizzes.

Math & Science Teacher **2014 – 2015**
Valiasr English Elementary School and Mesbah English Elementary School, Tehran, Iran

Private Tutoring — Mathematics and Physics **Summer 2012 and 2012 – 2014**

- Private lectures on freshman-level calculus and physics for high school and university students.

Lecture on Successful University Admission **Summer 2012**
Erfan High School and Beheshti High School, Tehran, Iran

- Delivered talks on pursuit of successful education and strategies for the National University Qualification Exam (Konkour) for women's high school audiences.

LEADERSHIP ROLES

Research Mentor **Jan 2022 – Jan 2023**
University of Colorado Boulder

- Trained new Ph.D. students on fundamentals of mechanics, written and oral skills, and technical research.
- Guided a research trainee through the modeling and simulation of beam bending in humidity actuators by application of the adsorptive poroelastic model.

President/Co-president **Aug 2018 – May 2019**
Persian Student Organization, Boulder, CO

- Proposed, secured funding, planned, and oversaw student events in collaboration with professionals from industry and academia, local businesses, and non-profit organizations.
- Proposed event plans and secured event funding with the University of Colorado Boulder Events Board.
- Managed teams of 5–10 members for event planning, advertisement, and execution.

TECHNICAL SKILLS

Programming & Software Development
Python, TypeScript/JavaScript, MATLAB, VBA, AutoLISP, FORTRAN, C++, HTML/CSS, SQL, PHP, Pascal

Platforms & Frameworks
Palantir Foundry (Ontology, Pipeline Builder, Workshop, OSDK, AIP, Spark, Polars, TypeScript/Python pipelines), Flask, Django, Palantir Foundry UI

Engineering Software
PLSCADD/PLS-TOWER, L-PILE, Autodesk AutoCAD, ABAQUS, LS-DYNA, ETABS, Plaxis, GeoStudio, SAP2000, SAFE, Civil3D, WaterGEMS, SewerCAD, StormCAD

Other Tools

Git, Microsoft Office Suite, LaTeX, Inkscape, Corel Draw, OriginPro

Engineering Standards
NESC, GO-95, FERC-881

LANGUAGE SKILLS

English — Fluent/Native
Persian/Farsi — Native

OTHER INTERESTS

- I am a plant hobbyist, animal lover, dog owner, and skilled chef for humans and dogs.
- I have a great passion for programming as a hobby.
- I enjoy cycling, swimming, jogging, and playing soccer.
- I play guitar and piano (beginner).
- I regularly watch movies and listen to music.